PROG7314 · PORTFOLIO OF EVIDENCE

PART 02



AEGIS

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For this project, AI tools (Claude AI and Gemini) were used in the following areas to research concepts and troubleshoot implementation issues. In each case the AI's output was reviewed and adapted before being used: understanding and final implementation decisions remained with the team.

REST API vs hosted API (Claude AI). Used to clarify the difference between a REST API (an architectural style resource URLs, HTTP verbs like GET/POST/PUT/PATCH/DELETE, statelessness, JSON responses with standard status codes) and a hosted API (one deployed and reachable at a public URL, such as on Render, rather than the only on localhost). This clarified why the team's Express backend needed to be deployed rather than run only on a developer's machine, since a real device or an APK can't reach localhost.

Firebase, Google Sign-In and Firestore setup (Claude AI). Used to plan the end-to-end Firebase setup: registering the Android app in the Firebase console against its applicationId, collecting debug-keystore SHA-1 via correctly wiring the Google Service Gradle plugin through the project's version catlog (including resolving a duplicate plugin declaration error), implementing Google Sign-in with the modern Credential Manger API (GetGoogleIdOption and GoogleAuthProvider, in place of the deprecated GoogleSignInClient), and writing production-mode Firebase security rules that restrict each user to reading and writing only their own document, keyed by uId.

Home screen layout (Gemini). Used to generate, then correct, the XML layout for the Volunteer Home feed screen: a ConstraintLayout header bar, a scrollable feed of MaterialCardView event cards showing status badges and attendance counts, and a BottomNavigationView with Home/Listings/Profile tabs after an initial response had formatting issues when copied into Android Studio.

Jetpack Datastore (Claude AI). Used to understand what Datastore is and why it replaces SharedPreferences for local settings storage: Preferences DataStore vs Proto DataStore, its asynchronous Coroutines/Flow-based reads and writes, transactional safety, and a worked example (a boolean preference key, a Flow-based read, and suspend function write). This directly informed how the project's local settings (dark mode, notifications, language, and the interest/skills selections for Custom feature 1) are persisted.

All cross all these conversations, AI was used to research and validate concepts, clarifying SDK setup steps, terminology and best practice.

AI chat links:

Claude AI: https://claude.ai/share/43e52ff9-9bc2-4e91-87e6-290f003bb0ca
Purpose: clarifies the difference between a REST API and a hosted API, and why the backend needed to be deployed rather than run locally.

Gemini: https://share.gemini.google/dUvdrz8i72SL
Purpose: generates and corrects the XML layout for the Volunteer Home feed screen (header, event cards, bottoms navigation).

Claude AI: https://claude.ai/share/9eb21fa7-7849-441e-b0e8-461abeddb40f
Purpose: walks through the end-to-end Firebase setup for Google Sign-In and Firestore, including SHA-1 fingerprints and security rules.

Claude AI: https://claude.ai/share/8d8f6725-7672-4392-9aae-9a84bf4efd41
Purpose: explains what Jetpack DataStore is and why it replaces SharedPreferences for local settings storage.